

Dynamic-routing MVP findings — proxy-through-gluetun (Mullvad) e2e

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Operator-authorized (2026-07-02): reset podman + prove the full proxy-through-gluetun Mullvad e2e. This document records the honest results (§11.4.6) — what is proven and the real MVP dynamic-routing defects that block the full squid-integrated path.

Proven live (rock-solid captured evidence)

- **podman/aardvark-dns is NOT broken** — throwaway inter-container DNS (10.89.2.2) and outbound DNS both resolve; the earlier "can't bind :53" fault has cleared. **No destructive podman system reset was done** — it would have destroyed the operator's 263 volumes (catalogizer/deploy/aosp/lava data — §9.2/§11.4.174).
- **squid-dynamic image rebuilt** (control-plane + squid multi-stage, config/squid context) → bakes http_port :34128 (E's §11.4.108 staleness RESOLVED; source was already 34128).
- **Dynamic stack boots; gluetun WireGuard tunnel to Mullvad is UP** (exits observed Osaka / Denver / Malmö — public IP is a Mullvad exit each boot).
- **CORE proxy-through-gluetun egress PROVEN:**
http_proxy=http://proxy-gluetun:8888 → am.i.mullvad.net/json = mullvad_exit_ip:true (194.114.136.116, Osaka). Evidence qa-results/dynamic_e2e_20260702T175536Z/.
- **Squid dynamic-routing LOGIC proven:** un-routed target → fail-closed TCP_DENIED/503 ERR_TUNNEL_DOWN (§11.4.68); with a route:<host:port> + up-status the acl-helper returns OK tag=<tunnel> and squid flips to TCP_TUNNEL (allowed). The decision engine works.

Open MVP dynamic-routing DEFECTS (the full squid-integrated path does NOT yet pass)

#	Defect	Status
D1	Compiler renders cache_peer gluetun- <profile> but the MVP ships one proxy-gluetun → peers unresolvable → all routed targets 503.	FIXED 068dc78 (network aliases on the gluetun compose service).
D2	Route key is port- qualified: squid %>ha{Host} sends host:443 on CONNECT, so routes must be route:<host>:<port>.	Documented (compiler/doc note).
D3	healthd↔gluetun health signal BROKEN: gluetun control API (/v1/ publicip/ip, /v1/vpn/ status) returns EMPTY; healthd's wg show <if> reads a WG interface in gluetun's netns (not healthd's) → every profile held falsely "down" → squid fail-closes even though the tunnel is genuinely UP.	FIXED 4a69225 (#79) + activated 2cd7f64. Took a THIRD path (neither doc option): healthd does a FRESH per-poll through-tunnel liveness PROBE via gluetun's built-in :8888 HTTP forward proxy — a real request egresses via the tunnel + returns THIS cycle (kill-switch blocks it when down ⇒ fail-closed). Additive to DecideHealth (§11.4.68- preserved, wg path intact), default- OFF until HEALTHD_TUNNEL_PROXY set. LIVE-PROVEN: the fixed healthd publishes state=up for the live Mullvad tunnel (qa-results/dynamic/ d3_healthd_live_20260702T200259Z/ + d3_liveprobe_...195836Z/).
D4	gluetun compose healthcheck probes /v1/openvpn/ status for a WireGuard tunnel → gluetun labelled "unhealthy" though the WG tunnel is up.	FIXED 57cc857: healthcheck → VPN- agnostic /v1/vpn/status + a read- only gluetun control-auth config (gluetun v3.40 made control routes private/401) so healthd + the healthcheck stop 401ing. Live- validated (gluetun healthy; status 200; mutating routes 401) + §11.4.135 guard.
D5	Squid caches the gluetun-* peer	OPEN.

#	Defect	Status
	negative DNS from its pre-alias startup; reconfigure did not clear it (peer stays dead). Needs squid to (re)resolve peers after gluetun is up (startup ordering / DNS lifecycle).	

Honest conclusion (§11.4.6)

The proxy-through-gluetun **egress capability** is proven live (gluetun's proxy egresses via Mullvad). **Rev 2 update (2026-07-02): D3 and D4 are now FIXED** (this session) — D3 (the deepest blocker, healthd false-down) resolved by the fresh through-tunnel liveness probe (#79, LIVE-PROVEN healthd publishes state=up) and D4 by the gluetun control-auth + VPN-agnostic healthcheck. The **full squid-integrated** path's LAST remaining blocker is **D5** (squid caches the gluetun-* peer negative-DNS from its pre-alias startup — peer stays dead until squid re-resolves after gluetun is up; a startup-ordering / DNS-lifecycle fix, tracked as a task). Until D5 is fixed, the full squid-integrated e2e is **NOT yet a passing e2e** — no PASS is claimed on it (§11.4.6). D1 fixed; D2 documented; D3+D4 FIXED; D5 open.